COMPUTER - NETWORKS : Why?

Tasks uses computer Networks.

- browsers
- Email - client
- Skype / whatsapp - call
- Remote access

Laptop
desktop
Smartphone
Smartwatch
→ computational device can use computer networks.

SDE, Bigdata Engineer, Network Engineer
↳ may end up doing networking.

* WIFI / LAN / 4G → These are devices used to communicate and technology. in networking

* Terms you will encounter while this course:

MAC address
IP-address
Router
WIFI
DNS
Subnet
HTTP / HTTPS

* Wireshark is one of the software used for the analyzing the scenario of incoming and outgoing traffic.

OSI - Model (Open System - Interconnect)

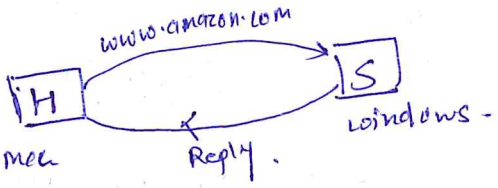
This model gives you the high level overview of how do you transmit data from one computer to another computer.

→ Reference - model.

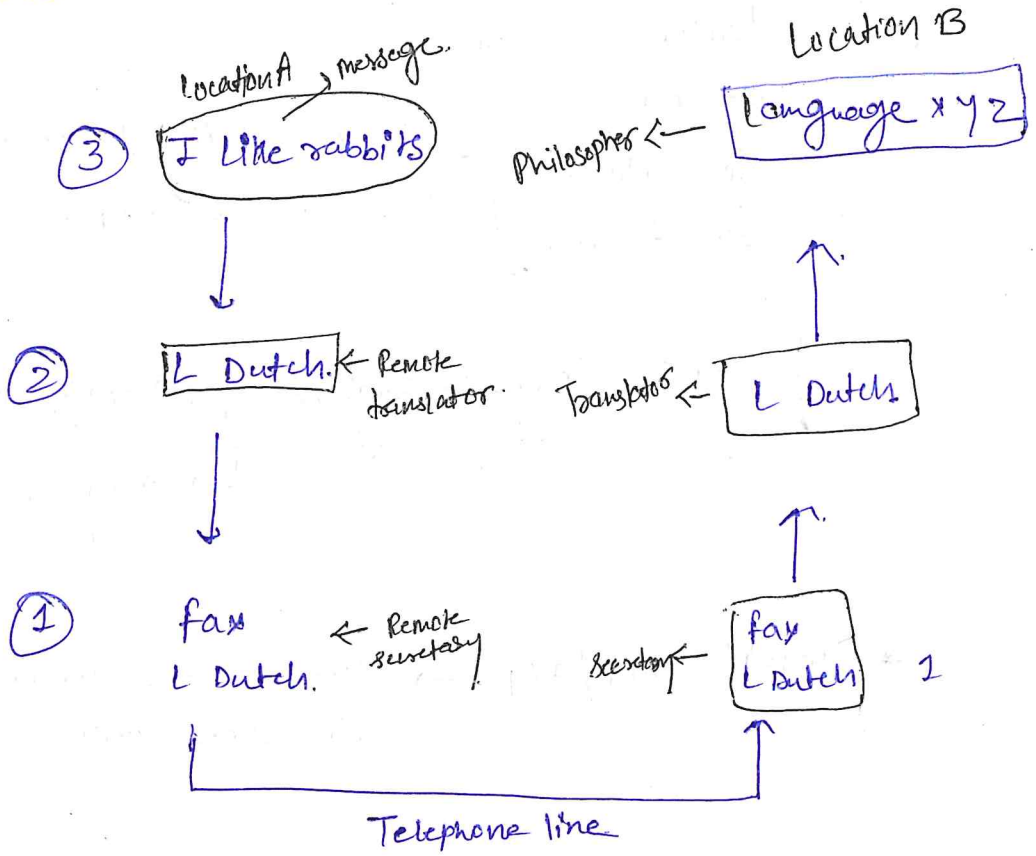
→ 1970's and 80's (1984)

→ Protocols (Rules) and Layered - system

→ host, address.



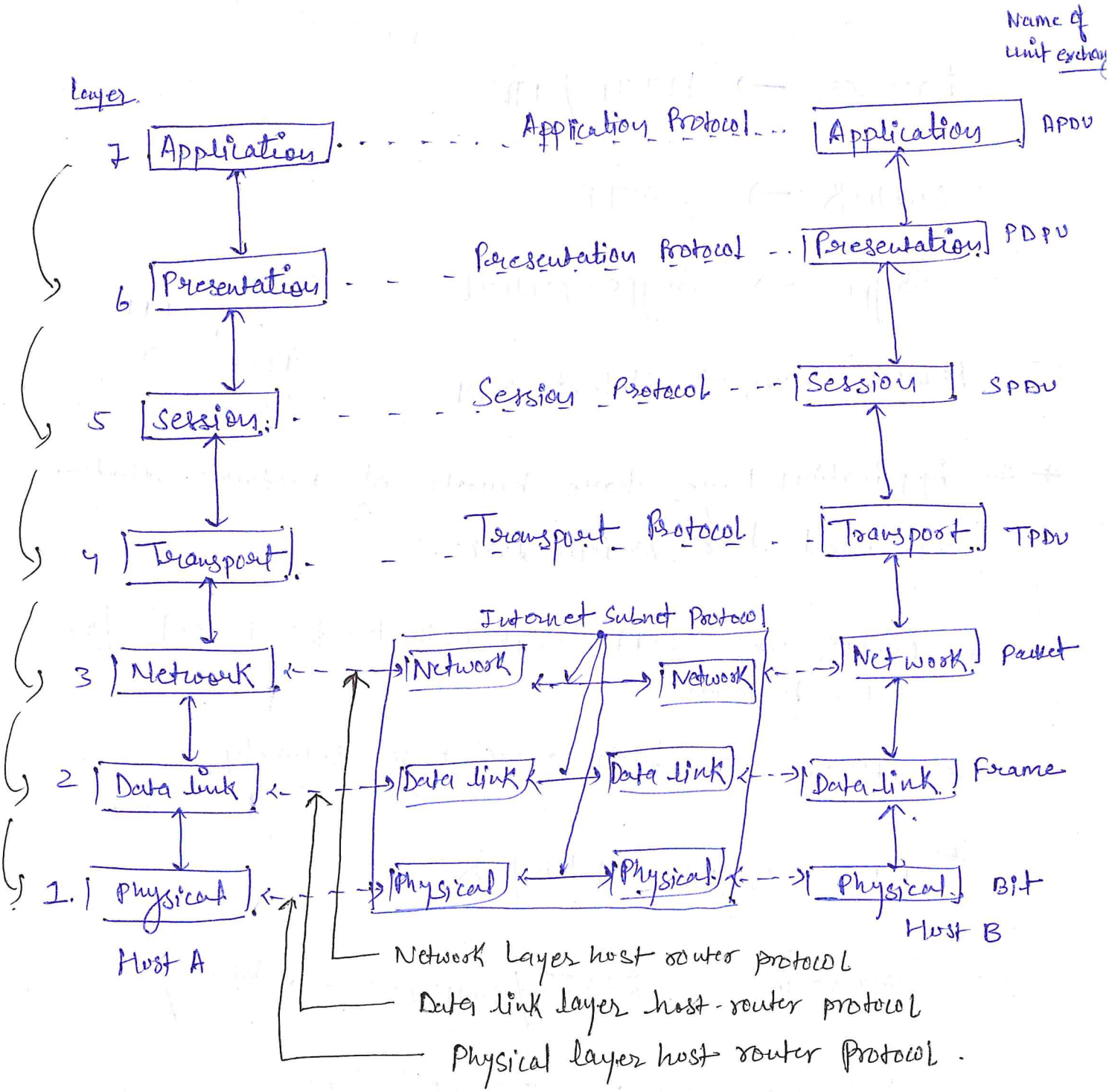
Layered approach →



* Co/os ⇒ address means Memory addresses.

* Computer networks ⇒ address means IP address / MAC address

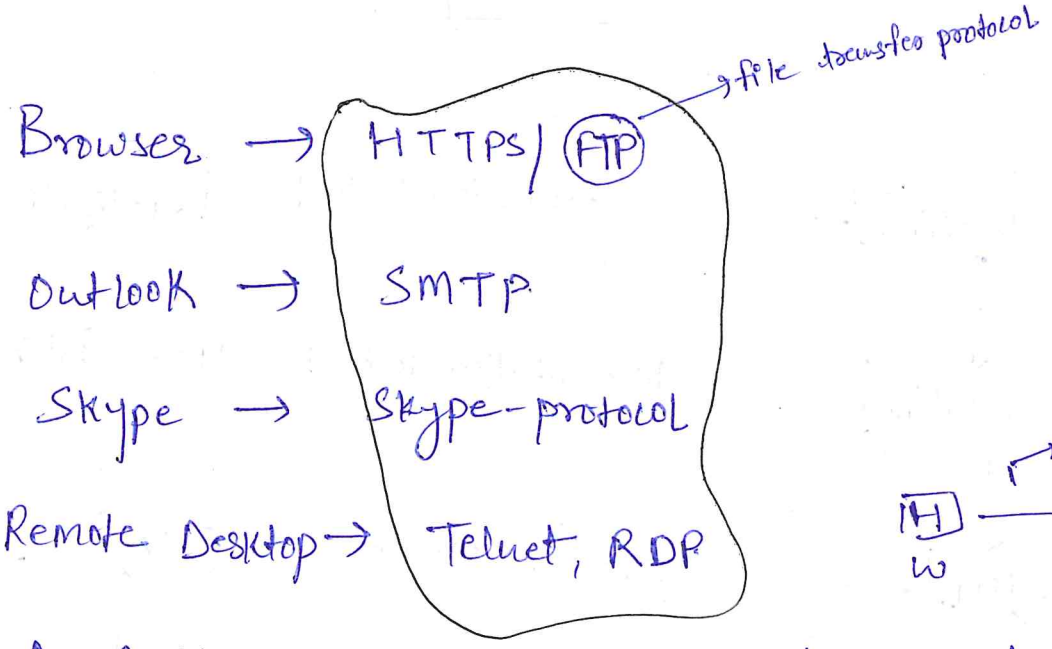
OSI Layers



OSI REFERENCE MODEL

→ In coming course we are going to learn about each of these layers in detail.

Application Layer :—



** Application Layer have bunch of protocols that are used for various tasks.

* Outlook uses SMTP (simple mail transfer protocol) for transfer mail.

* HTTPS to fetch a web page securely.

* PRESENTATION LAYER :—

→ Translation ASCII / UNICODE → binary

→ Data compression Text / Audio / video

→ Encryption Passwords

SSL (Secure Socket Layer)

DS and Algo (HuffmanAlgo)

→ Translate one form to another form.

→ Compress the data from one form to another form.

→ Encrypt the data from one form to another form.

Session Layer :—

→ Establish, manage and terminate connections.

→ Authorization and Authentication.

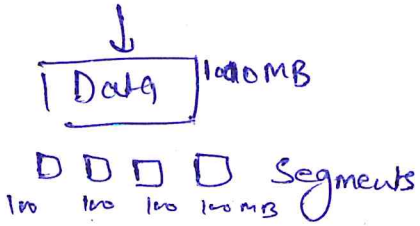
* Example of session is login and logout.

* Authentication using userid and password.

Most modern browsers manages these tasks → { Application, Presentation, Session Layer }

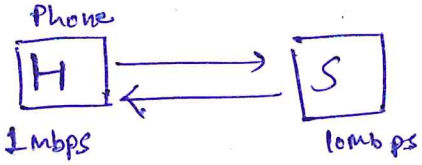
Transport Layer :—

→ Segmentation



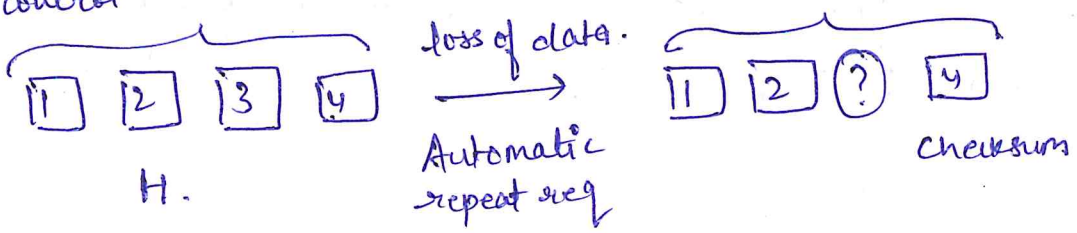
* Data is breaking into segments to transfer one chunk of data

→ FLOW - control



* At what speed the host wants the flow from the server.

→ Error control



There can be lot of errors while transmitting the data. Error control mechanism helps us to detect and correct the errors. We use checksum, CRC etc.

*

* Port no. 80 used widely for browsers.

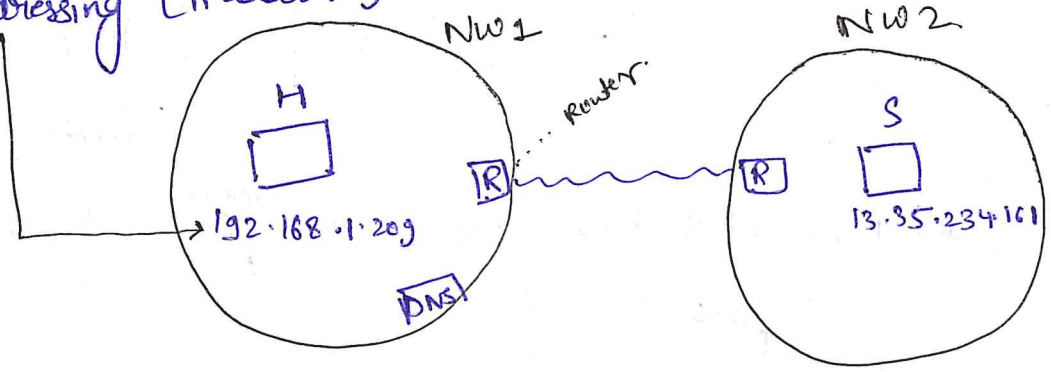
→ TCP VS UDP.

Slow	fast, lossy
lossless	no feedback
feedback	videochat
webpage, email	
Connection oriented	Connection less.

Network Layer :-

→ Packets [Segment + Source and destination address] ⇒ We include Ip addresses in the Tcp packet.

→ Logical addressing (Ip address)



* Range of ip address: 0.0.0.0 to 255.255.255.255
* You can use wireshark to capture the packets and also you can know various types of packets and lot of details about them.
* The task of routing is to route the packet from source to destination.

→ Routing Mask.

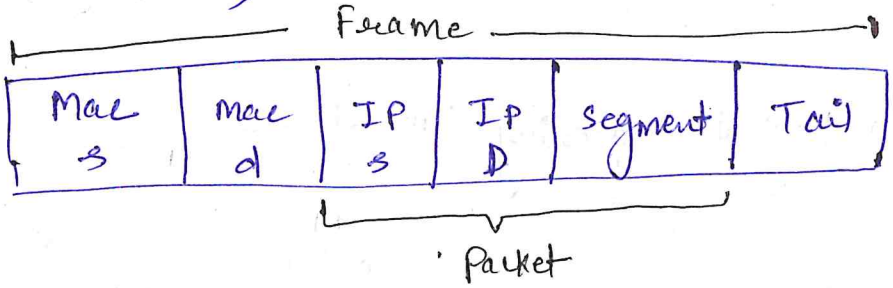
* The mask number help to get the network address by performing AND operation.

* DNS (Domain name system) is used to connect host name like amazon.com to the ip address.

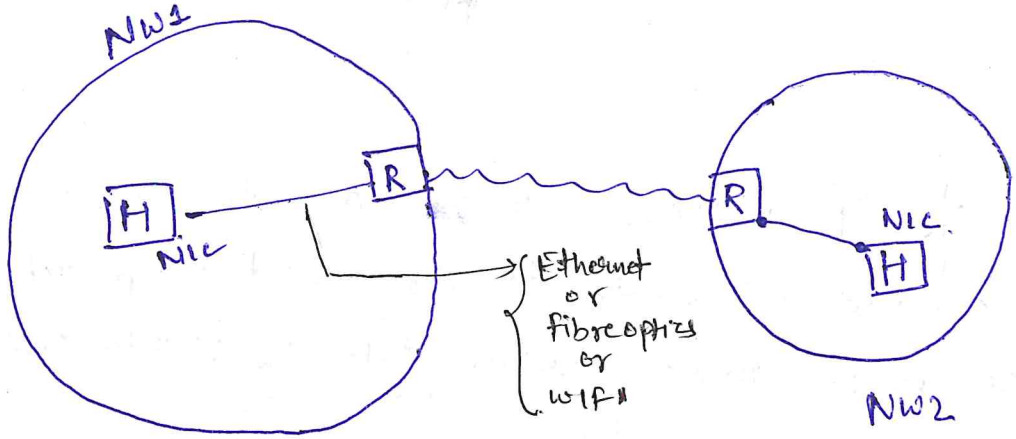
→ Path determination: To find most appropiate paths. [Graph theory application]

Data Link layer :- (L3)

→ Physical-addr (MAC)



* Data link layer we called data. of a frame

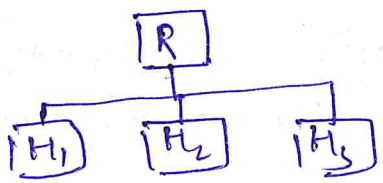


* Every host has a unique mac address.

* Data link layer added mac addresses.

→ Access to media (upper wire, fibre-optical-wire, wireless)
Ethernet

→ (MAC) Media-access-control : collision free



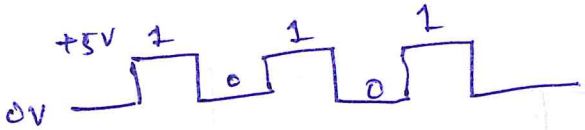
* All of the host can not send packet at same time. MAC ensures avoid collission

→ Error detection : Tail

It is added in the tail part like checksum.

Physical layer :—

→ Bits to signals



Copper wire

Laser - ON | off
fiber-optics

radio signals
wireless

Physical layer job is to transmit from bits to signal based on the medium you used.

1984
Reference model [Theoritical]

OSI

Application → DNS, HTTP | S
Presentation
Session
} Grouped together → Application → Data.

Transport → Transport → Segments
Network. → Network. → Packets
Data link.
Physical
} → Physical → Frames / Bitsream. (stream of bits.)

TCP/IP (practical)

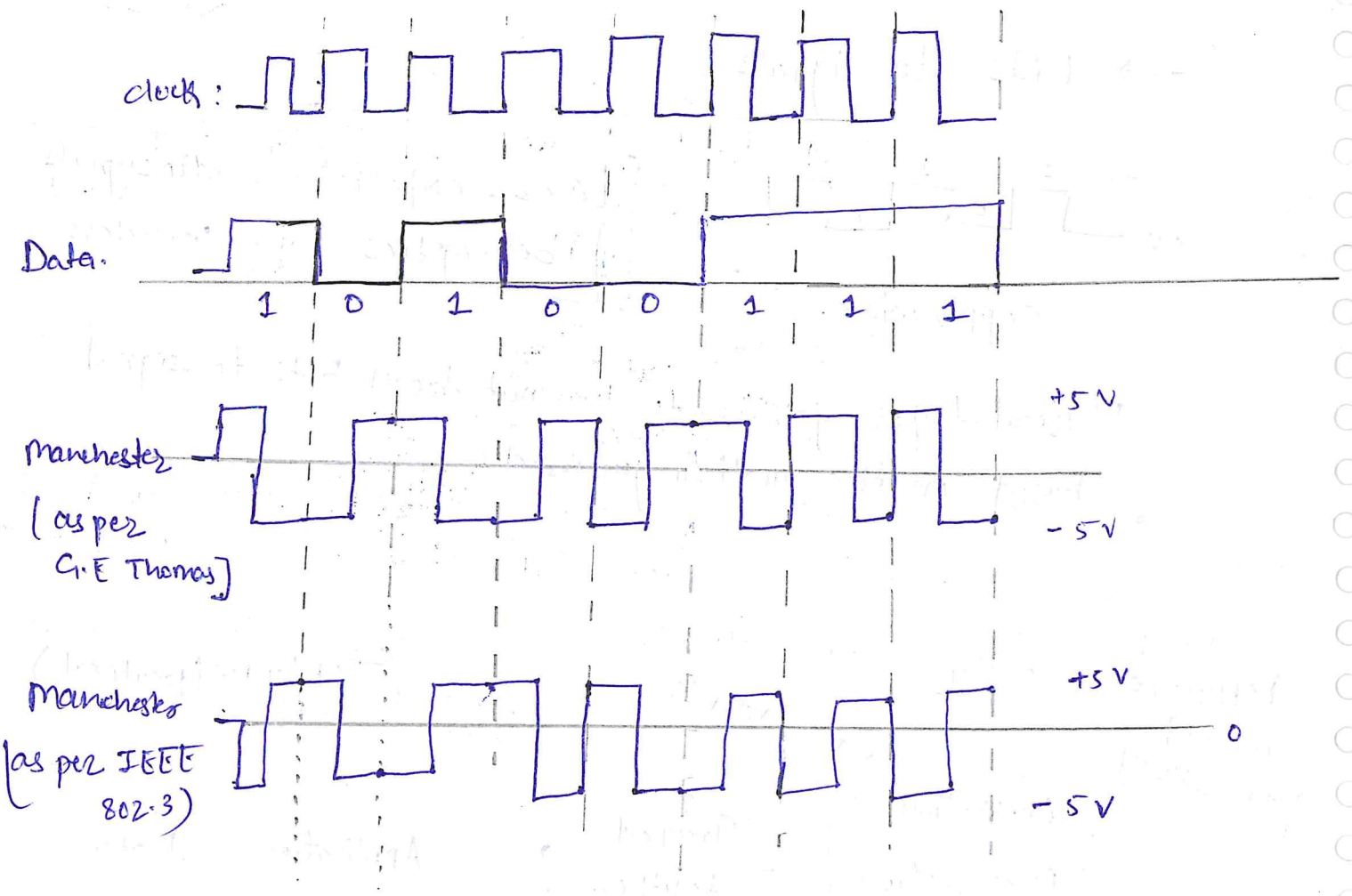
TCP/IP

* TCP/IP widely used practical model while OSI is Ref model.
* Segments are also called datagrams.

Physical Layer: —

Bits to signals.

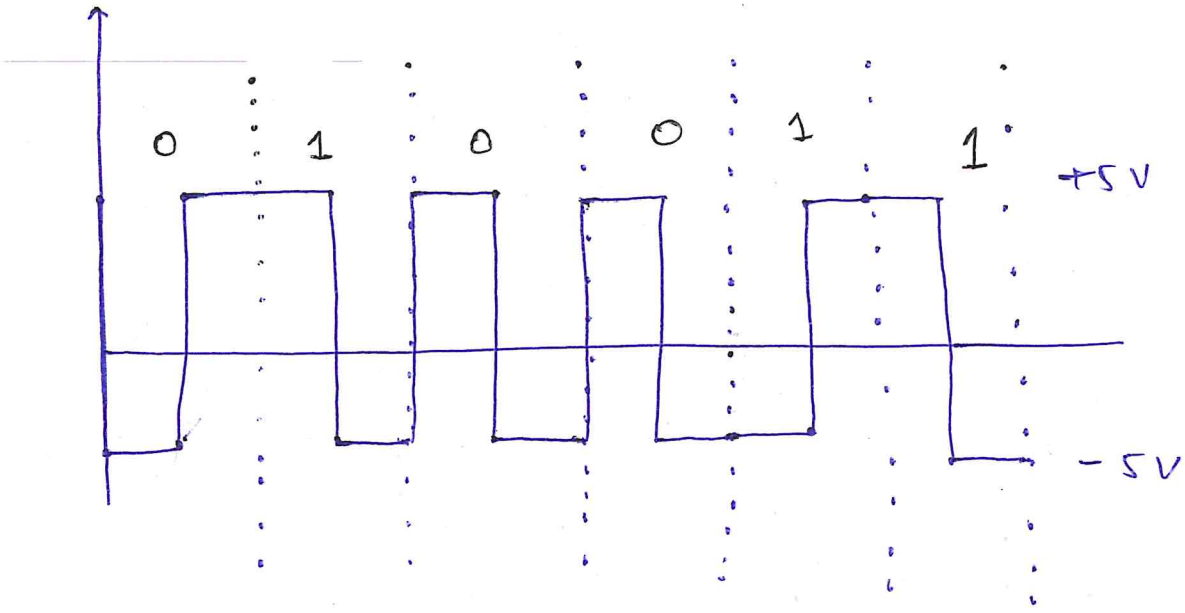
Encoding: — * You have each clock cycle.



GE Thomsan follows +A → 1, A → 0, -A

IEE 802.3 → +A → 1, -A; +A → 0, -A

Differential manchester encoding



In differential manchester encoding, the transition at the middle of the bit is used only for synchronisation.

The bit representation is defined by the inversion.

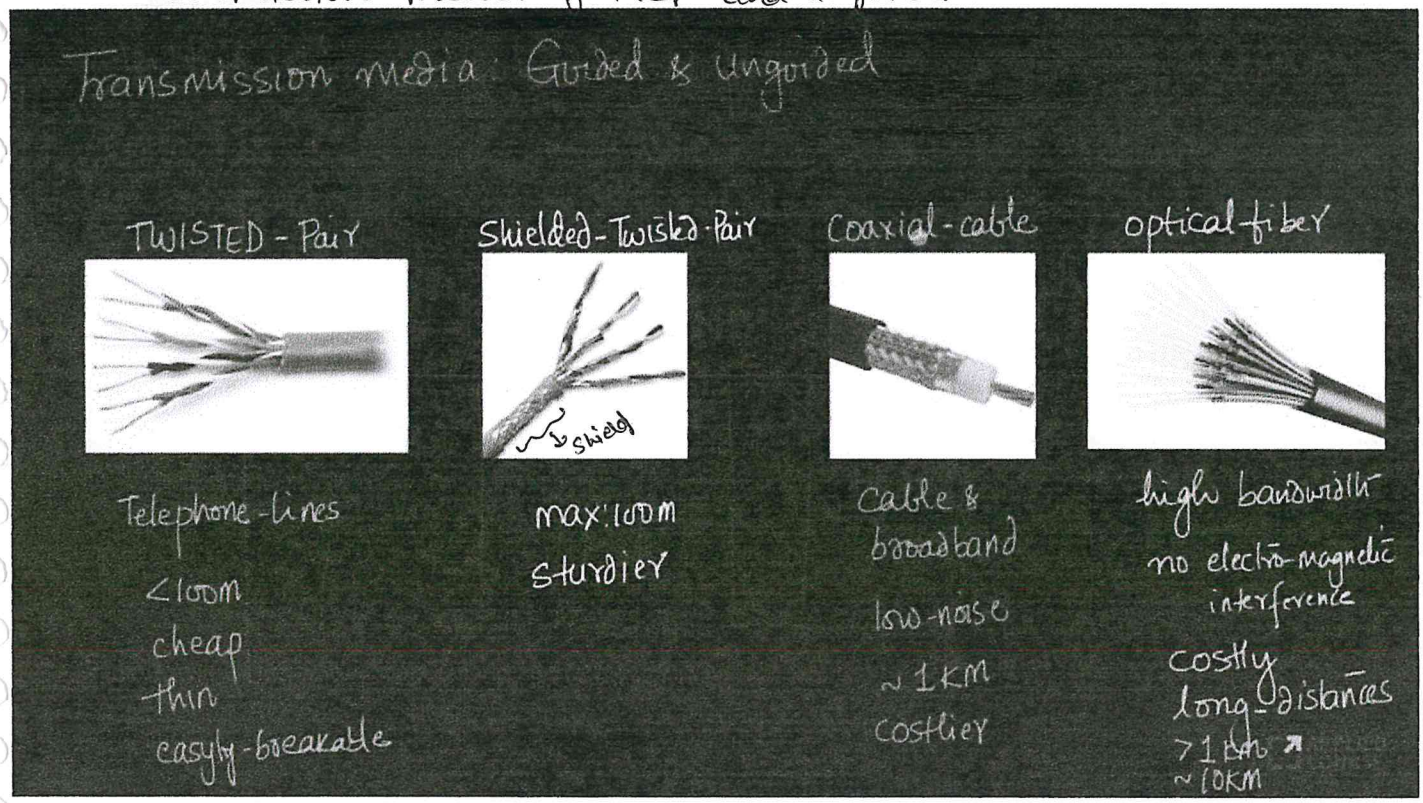
Bit 0 → Inversion

Bit 1 → No inversion.

Whenever 0 arrives we invert the transition whenever 1 arrives we don't invert and continue the transition.

* In diff manchester encoding transition is from -v to +v or +v to -v. We are using + and - because manchester / diff manchester are bipolar encoding.

Transmission media guided and unguided :-



All twisted pair, Shielded - Twisted pair, co-axial cable and optical fiber are guided media.

Twisted pair → Simplest media, invented by Alexander grahmbell Modem connect to telephone lines and also to computer for internet. Here pair of copper wire twisted amongst themselves.

Shileded Twisted pair → They are shielded with aluminium shield. It makes it strengthful and also less interference.

Co-axial cable → Cable connection (DTH) used this most. It is costlier and work 1 KM distance. We can use the repeater to strength the cable.

Optical fiber → The laser beam is pass through it.
It uses the concept of total Internal reflection (TIR). It can work upto 10KM

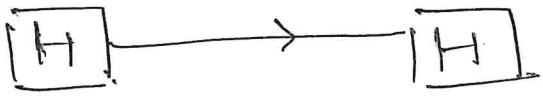
UNGuided media :-

Infrared
↓
* TV remotes uses this technology.
* Short range can't penetrate wall.
* Early 2010-2012 smartphones also used it.

wireless
↓
* radio-waves (WiFi, 4G)
(weather-condition)
↳ 15 KM.

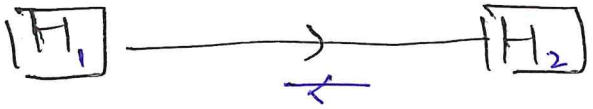
Transmission modes: -

Simplex →



In Simplex we can send the data only one direction.

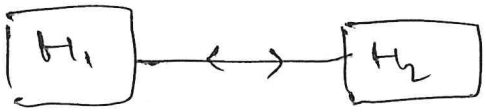
Half duplex →



In duplex we can send the data in both the directions but not at the same time.

Either host 1 to host 2.
or host 2 to host 1.

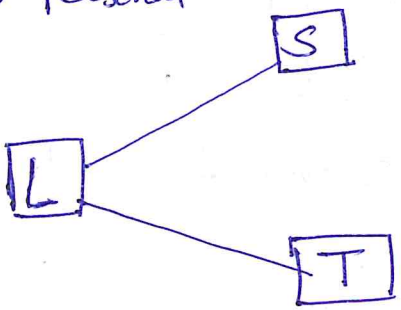
Full duplex →



In full duplex we can send the data in both the directions. we can also send the data in both the directions at the same time.

Network Types :-

① PAN → Personal area network.

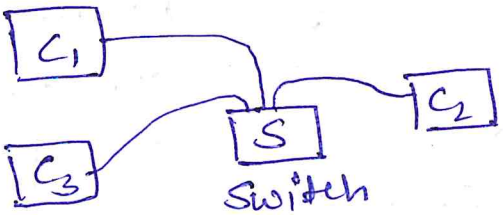


USB - cable
Bluetooth → most popular
NFC → Payment systems.
Infrared ← IR

NFC → Nearfield communication

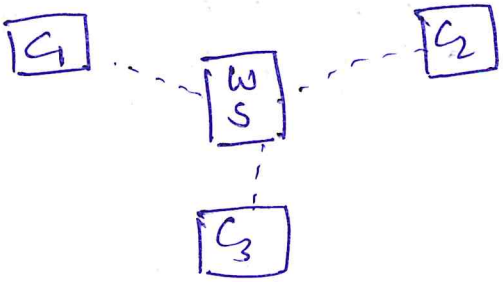
* A network that you can construct using the personal devices without use of infrastructure.

② LAN [Ethernet-LAN]



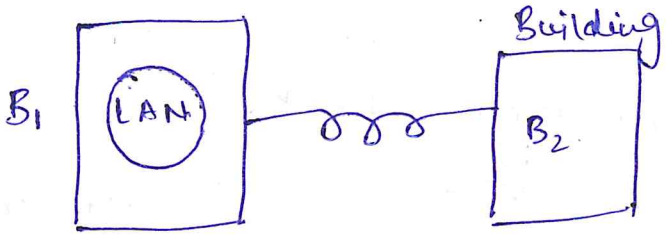
Local area network. uses networking devices like hub, switch, bridges etc.

③ WLAN (WIFI)



* The repeater gives the boost to the signal and transmits it further.

④ CAN (Campus Area Network)



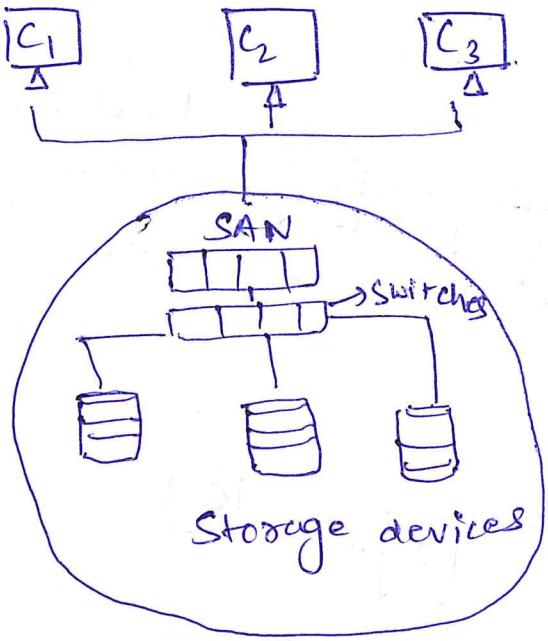
Connection at the campus level is called CAN.

⑤ MAN → Level of Town.

⑥ WAN → country, Europe, world

eg: Internet

⑦ SAN [Storage area network] :-



All the systems are connected to very fast switches and router. The transfer of data from these storage devices.